

Day 3 — TNPSC Group 2A Study Material

Part A – General Studies: Mechanics — Properties of Matter, Force, Motion, and Energy

READING MATERIAL

This is the densest physics-fact clause in Unit I — TNPSC tests it through a mix of motion-type identification, formula-matching ('which of these equations is correct'), and real-world application questions (why X happens) rather than asking for bare definitions in isolation.

Q: If the distance travelled by a body is directly proportional to the square of time, what type of motion does it have?

A: Uniformly accelerated motion. (Distance $\propto t^2$ is exactly the shape of the equation $s = \frac{1}{2}at^2$ for motion starting from rest under constant acceleration.)

Q: Which of these basic mechanics equations are correctly stated? (i) Force = mass $\times$ velocity (ii) Acceleration = velocity/time (iii) Work = Force $\times$ displacement (iv) Power = Work/time

A: (ii), (iii) and (iv) are correct. (i) is wrong — Force = mass $\times$ ACCELERATION ($F=ma$), not mass $\times$ velocity (mass $\times$ velocity is actually the formula for momentum).

Q: In uniform circular motion, what is linear velocity always perpendicular to?

A: The radius vector (the line from the centre to the moving object) — this is why circular motion requires a constant centre-directed (centripetal) force even though speed stays constant.

Q: Why do rear-wheel-drive cars (engine in front) often need extra weight in the trunk, especially in one driving condition?

A: On icy roads — added weight over the rear (drive) wheels improves traction/grip, which is especially important on low-friction icy surfaces.

Q: Why is a water drop in space (near-zero gravity) spherical?

A: In the absence/near-absence of gravity, surface tension dominates, and a sphere is the shape with the least surface area for a given volume — so the drop naturally settles into a sphere to minimize surface energy.

Q: What are Newton's three laws of motion, in brief?

A: First law (inertia): a body stays at rest or in uniform motion unless acted on by a net external force. Second law: $F = ma$ (force equals mass times acceleration). Third law: for every action, there is an equal and opposite reaction.

Q: What's the difference between kinetic energy and potential energy, with their formulas?

A: Kinetic energy (KE) is the energy of motion: $KE = \frac{1}{2}mv^2$. Potential energy (PE) is stored energy due to position (commonly gravitational): $PE = mgh$ (mass $\times$ gravity $\times$ height).

Q: What is momentum, and how is it different from force?

A: Momentum = mass $\times$ velocity ($p = mv$) — a property of a moving object. Force = mass $\times$ acceleration ($F = ma$) — what's needed to change that motion. A common wrong-answer trap is swapping these two formulas.

MOCK TEST (WITH ANSWERS)

1. A body's distance travelled is directly proportional to the square of elapsed time. This indicates:

- A. Uniform velocity
- B. Uniformly decelerated motion
- C. Uniformly accelerated motion ✓**
- D. No motion

Explanation: $s \propto t^2$ matches $s = \frac{1}{2}at^2$, the signature of uniformly accelerated motion from rest. [medium, web-research]

2. Which of the following equations is INCORRECT?

- A. Acceleration = velocity/time
- B. Force = mass $\times$ velocity ✓**
- C. Work = Force $\times$ displacement
- D. Power = Work/time

Explanation: Force = mass $\times$ acceleration, not mass $\times$ velocity (that's momentum). [medium, web-research]

3. In uniform circular motion, the linear velocity at any instant is perpendicular to the:

- A. Force
- B. Momentum
- C. Radius vector ✓**
- D. Mass

Explanation: Perpendicular to the radius vector — this is the geometric basis of circular motion. [medium, web-research]

4. Why is extra weight sometimes added to the trunk of a rear-wheel-drive car when driving on icy roads?

- A. To reduce speed
- B. To improve traction on the drive wheels ✓**
- C. To improve fuel economy
- D. To improve visibility

Explanation: Extra weight over the rear (drive) wheels improves grip/traction on low-friction icy surfaces. [medium, web-research]

5. A water droplet floating in zero gravity (e.g. in space) takes a spherical shape because a sphere:

- A. Has maximum surface area for its volume
- B. Has minimum surface area for its volume ✓**
- C. Has maximum volume for its surface area
- D. Is the easiest shape to photograph

Explanation: Minimum surface area for a given volume minimizes surface energy, which surface tension favors. [medium, web-research]

6. Newton's Third Law states that:

- A. A body at rest stays at rest unless acted on by a force
- B. Force equals mass times acceleration
- C. For every action there is an equal and opposite reaction ✓**
- D. Energy can neither be created nor destroyed

Explanation: Newton's Third Law — action-reaction pairs. [easy, authored]

7. The formula for kinetic energy of a moving body is:

- A. mgh
- B. $\frac{1}{2}mv^2$ ✓**
- C. mv
- D. ma

Explanation: $KE = \frac{1}{2}mv^2$. [easy, authored]

8. The formula for gravitational potential energy is:

- A. $\frac{1}{2}mv^2$
- B. mgh ✓**
- C. mv
- D. ma

Explanation: $PE = mgh$ (mass $\times$ gravitational acceleration $\times$ height). [easy, authored]

9. Momentum of a moving body is given by:

- A. mass $\times$ acceleration
- B. mass $\times$ velocity ✓**
- C. force $\times$ time
- D. $\frac{1}{2} \times$ mass $\times$ velocity²

Explanation: Momentum $p = mv$. [easy, authored]

10. Which of Newton's laws is also known as the 'law of inertia'?

- A. First law ✓**
- B. Second law
- C. Third law
- D. None of these

Explanation: The First Law (a body resists changes to its state of motion) is the law of inertia. [easy, authored]

Part B – Aptitude & Mental Ability: Highest Common Factor (HCF)

READING MATERIAL

HCF is 1 of 10 named Aptitude sub-skills. Real exam evidence (4 confirmed hits across 2022/2024/2025) shows TNPSC favors HCF *application* problems — measuring/tiling/container word problems and the HCF-LCM relationship — over plain 'find the HCF of these two numbers' questions, so the reading below leads with those application shapes.

Q: Real exam (2025): Two numbers are in the ratio 15:11. If their HCF is 13, find the numbers.

A: When two numbers are in a given ratio, write them as $15k$ and $11k$. Since $\text{HCF}(15,11)=1$ (the ratio is already in lowest terms), k IS the HCF — so $k=13$. The numbers are $15 \times 13=195$ and $11 \times 13=143$.

Q: Real exam (2025): What is the least number of square tiles needed to pave the ceiling of a room 15m 17cm long and 9m 2cm broad?

A: Convert to one unit: $1517 \text{ cm} \times 902 \text{ cm}$. The LARGEST square tile that exactly fits (no cutting) has a side equal to the HCF of the two dimensions. $\text{HCF}(1517, 902) = 41 \text{ cm}$ (via the Euclidean algorithm: $1517=1 \times 902+615$; $902=1 \times 615+287$; $615=2 \times 287+41$; $287=7 \times 41+0$). Tile area = $41 \times 41 = 1,681 \text{ cm}^2$. Room area = $1517 \times 902 = 1,368,334 \text{ cm}^2$. Number of tiles = $1,368,334 \div 1,681 = 814$.

Q: Real exam (2024): If the HCF of x^3-8y^3 and $x^2-k^2y^2$ is $x+ky$, find k .

A: Factor x^3-8y^3 using $a^3-b^3=(a-b)(a^2+ab+b^2)$ with $a=x$, $b=2y$: $x^3-8y^3 = (x-2y)(x^2+2xy+4y^2)$. Factor $x^2-k^2y^2$ using $a^2-b^2=(a-b)(a+b)$: $x^2-k^2y^2 = (x-ky)(x+ky)$. For ' $x+ky$ ' to be the common factor (HCF) of both, it must match one of the first expression's factors. $(x-2y)$ is the only linear factor available there, so $x+ky$ must equal $x-2y$, giving $k=-2$.

Q: Real exam (2022): What is the greatest possible volume of a vessel that can exactly measure the milk in cans of 80, 100, and 120 litres (each can measured to full capacity only)?

A: This is asking for the HCF of 80, 100, and 120. $\text{HCF}(80,100)=20$, then $\text{HCF}(20,120)=20$. So the answer is 20 litres — the largest vessel that can measure each can's full volume an exact whole number of times.

Q: Real exam (2022): The HCF and LCM of two numbers are 16 and 240 respectively. If one number is 48, find the other.

A: Use the relationship: $\text{HCF} \times \text{LCM} = \text{product of the two numbers}$. $16 \times 240 = 3,840$. Other number = $3,840 \div 48 = 80$.

Q: Method — finding HCF via the Euclidean algorithm (the fast way for large numbers): What is the HCF of 1517 and 902?

A: Repeatedly divide the larger by the smaller and replace the larger with the remainder, until the remainder is 0 — the last nonzero remainder is the HCF. $1517=1 \times 902+615 \rightarrow 902=1 \times 615+287 \rightarrow 615=2 \times 287+41 \rightarrow 287=7 \times 41+0$. $\text{HCF}=41$. This is the same calculation used in ro2 above — recognize it as the standard fast method rather than listing all factors of both numbers.

MOCK TEST (WITH ANSWERS)

1. Two numbers are in the ratio 21:14. If their HCF is 12, find the larger number.

A. 36 ✓

B. 24

C. 252

D. 168

Explanation: Watch out: 21 and 14 are NOT coprime ($\text{gcd}=7$), so you can't directly multiply them by the HCF. Reduce the ratio first: $21:14 = 3:2$ (coprime). Now apply the HCF: larger = $3 \times 12=36$, smaller = $2 \times 12=24$. [hard, authored]

2. Find the largest square tile that can exactly pave a floor measuring 3m 8cm by 4m 20cm (no cutting).

A. 28 cm ✓

B. 14 cm

C. 56 cm

D. 84 cm

Explanation: Convert to cm: 308 and 420. $\text{HCF}(308,420)$: $420=1 \times 308+112$; $308=2 \times 112+84$; $112=1 \times 84+28$; $84=3 \times 28+0$. $\text{HCF}=28 \text{ cm}$. [hard, authored]

3. If the HCF of a^2-4b^2 and $a^2+ab-6b^2$ is $a-2b$, this is consistent with which factorization of the second expression?

A. $(a+3b)(a-2b)$ ✓

B. $(a-3b)(a+2b)$

C. $(a+3b)(a+2b)$

D. $(a-3b)(a-2b)$

Explanation: $a^2-4b^2=(a-2b)(a+2b)$. The second expression factors as $(a+3b)(a-2b) = a^2+ab-6b^2$ — sharing the common factor $(a-2b)$, so that's the HCF. [hard, authored]

4. What is the greatest volume that can exactly measure the contents of drums holding 72, 96, and 120 litres?

A. 12 litres

B. 24 litres ✓

C. 8 litres

D. 6 litres

Explanation: $\text{HCF}(72, 96) = 24$, then $\text{HCF}(24, 120) = 24$. Answer: 24 litres. [medium, authored]

5. The HCF and LCM of two numbers are 12 and 180 respectively. If one number is 36, find the other.

A. 48

B. 54

C. 60 ✓

D. 72

Explanation: $\text{HCF} \times \text{LCM} = 12 \times 180 = 2,160$. Other number = $2,160 \div 36 = 60$. [medium, authored]

6. Using the Euclidean algorithm, find the HCF of 1,073 and 925.

A. 37 ✓

B. 29

C. 31

D. 43

Explanation: $1073 = 1 \times 925 + 148$; $925 = 6 \times 148 + 37$; $148 = 4 \times 37 + 0$. $\text{HCF} = 37$. [medium, authored]

7. Two numbers are in the ratio 8:9 and their HCF is 7. Find their sum.

A. 105

B. 119 ✓

C. 126

D. 112

Explanation: Numbers: $8 \times 7 = 56$ and $9 \times 7 = 63$. Sum = $56 + 63 = 119$. [medium, authored]

Part C – General English: Poems: "Teamwork" & "Courage" — both by Edgar Albert Guest

READING MATERIAL

Both poems are short, didactic verse by the same American poet (Edgar Albert Guest, also responsible for "The Stick-Together Families" elsewhere in this syllabus). Real exam evidence shows TNPSC tests these almost entirely via 'identify the figure of speech in this exact line' and 'what does the pronoun it/this refer to' — so the Q&A below drills both formats directly.

Q: What is the central message of "Courage"?

A: The poem defines courage by first saying what it ISN'T (not a brilliant dash, not instantaneous, not born of despair, not a creature of flickered hope), then says what it IS: something deep in the soul of man, always working to serve a plan, present in both victory and defeat, and in every trial a person faces. This 'define by negation, then affirmation' structure is itself worth recognizing for appreciation-style questions.

Q: Real exam line: "It's written in victory and defeat, And every trial a man may meet." What does 'it' refer to?

A: Courage — this exact line, with this exact pronoun-reference question, appeared in a real TNPSC paper (2024).

Q: Identify the figure of speech: "Courage isn't a brilliant dash" and "But it's something deep in the soul of man."

A: Both lines are cited as Metaphor — courage is described AS a 'dash' / AS something physically 'deep' inside a person, without using 'like/as'.

Q: Identify the figure of speech: "Born of despair with a sudden spring" and "It isn't a creature of flickered hope" and "That is working always to serve some plan."

A: All three are cited as Personification — courage (an abstract idea) is given creature-like or human-like qualities (being 'born', being a 'creature', 'working' toward a 'plan').

Q: What is the central message of "Teamwork"?

A: Individual fame and being 'singled out' for glory matter less than contributing to the team — 'there's no lone hand in the game we play.' True worth is shown by how you 'pull with the team' and work with your 'running mate' toward a 'bigger scheme', not by solo achievement.

Q: Real exam line: "It's helping your fellowman to score, When his chances hopeless seem." What does 'it' refer to?

A: Teamwork — this exact line, with this exact pronoun-reference question, appeared in a real TNPSC paper (2024).

Q: Identify the figure of speech: "And it's fine to be counted a star" and "For there's no lone hand in the game we play" and "We must work to a bigger scheme."

A: All three are cited as Metaphor — being 'a star', having a 'lone hand' (a card-game image for going it alone), and a 'bigger scheme' are all non-literal stand-ins for the poem's real subject (individual glory vs. teamwork).

Q: Identify the figure of speech: "It's all very well to have courage and skill" and "And the thing that counts in the world to-day."

A: Both lines are cited as Alliteration in the source material.

Q: Identify the figure of speech: "How do you pull with the team?"

A: Synecdoche — 'pull' (a single physical action, as in rowing or tug-of-war) stands in for a person's entire contribution and effort within the team.

Q: Identify the figure of speech: "They may single you out for fame."

A: Irony — cited as ironic given the poem's overall argument that individual fame/recognition is less meaningful than team contribution; being 'singled out' runs against the poem's central value.

MOCK TEST (WITH ANSWERS)

1. "It's written in victory and defeat, And every trial a man may meet." The word 'it' in this real exam line refers to:

- A. Hope
- B. Sports
- C. Courage ✓**
- D. Patience

Explanation: Courage — confirmed real 2024 TNPSC question. [medium, question-bank-2024]

2. "It's helping your fellowman to score, When his chances hopeless seem." The word 'it' in this real exam line refers to:

- A. Hard work

- B. Best work
- C. Smart work
- D. Team work ✓**

Explanation: Teamwork — confirmed real 2024 TNPSC question. [medium, question-bank-2024]

3. Which figure of speech is used in "Courage isn't a brilliant dash"?

- A. Simile
- B. Metaphor ✓**
- C. Personification
- D. Alliteration

Explanation: Metaphor. [medium, web-research]

4. Which figure of speech is used in "Born of despair with a sudden spring"?

- A. Simile
- B. Metaphor
- C. Personification ✓**
- D. Alliteration

Explanation: Personification — courage is described as being 'born'. [medium, web-research]

5. Which figure of speech is used in "It isn't a creature of flickered hope"?

- A. Hyperbole
- B. Simile
- C. Metaphor
- D. Personification ✓**

Explanation: Personification — courage is called a 'creature'. [medium, web-research]

6. Which figure of speech is used in "For there's no lone hand in the game we play" (from Teamwork)?

- A. Simile
- B. Metaphor ✓**
- C. Irony
- D. Synecdoche

Explanation: Metaphor — 'lone hand' (a card-game image) stands for going it alone. [medium, web-research]

7. Which figure of speech is used in "How do you pull with the team?"

- A. Simile
- B. Metaphor
- C. Irony
- D. Synecdoche ✓**

Explanation: Synecdoche — 'pull' represents the whole of a person's team contribution. [hard, web-research]

8. Which figure of speech is used in "They may single you out for fame"?

- A. Irony ✓**
- B. Metaphor
- C. Hyperbole
- D. Simile

Explanation: Irony — being singled out for individual fame runs against the poem's team-first message. [hard, web-research]

9. "And it's fine to be counted a star" uses which figure of speech?

- A. Simile
- B. Metaphor ✓**
- C. Hyperbole
- D. Alliteration

Explanation: Metaphor — 'a star' stands for being exceptionally accomplished. [medium, web-research]

10. Both "Teamwork" and "Courage" are written by:

- A. Robert Frost
- B. Edgar Albert Guest ✓**

C. Henry Van Dyke

D. Rudyard Kipling

Explanation: Edgar Albert Guest wrote both — and also 'The Stick-Together Families' elsewhere in this syllabus. [easy, web-research]